



Building Spark Application Using Scala/SBT

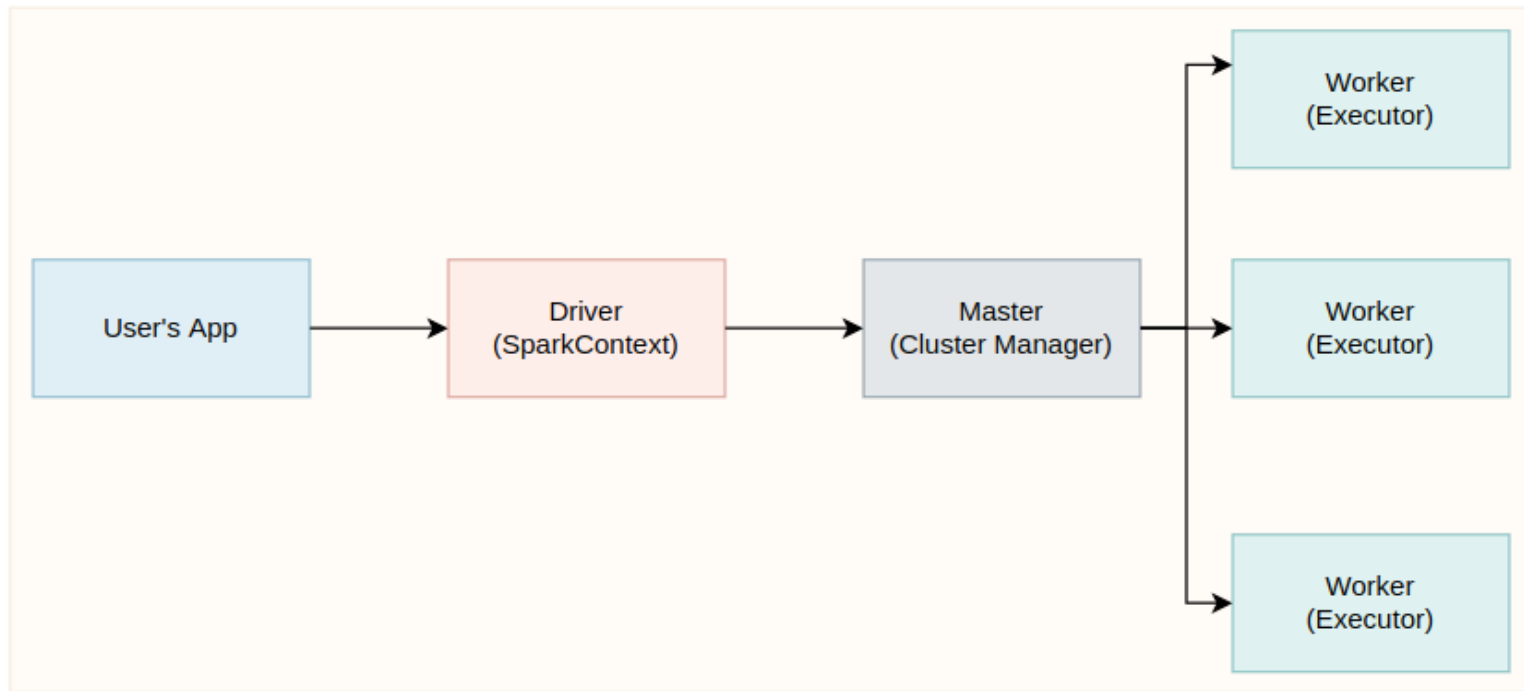
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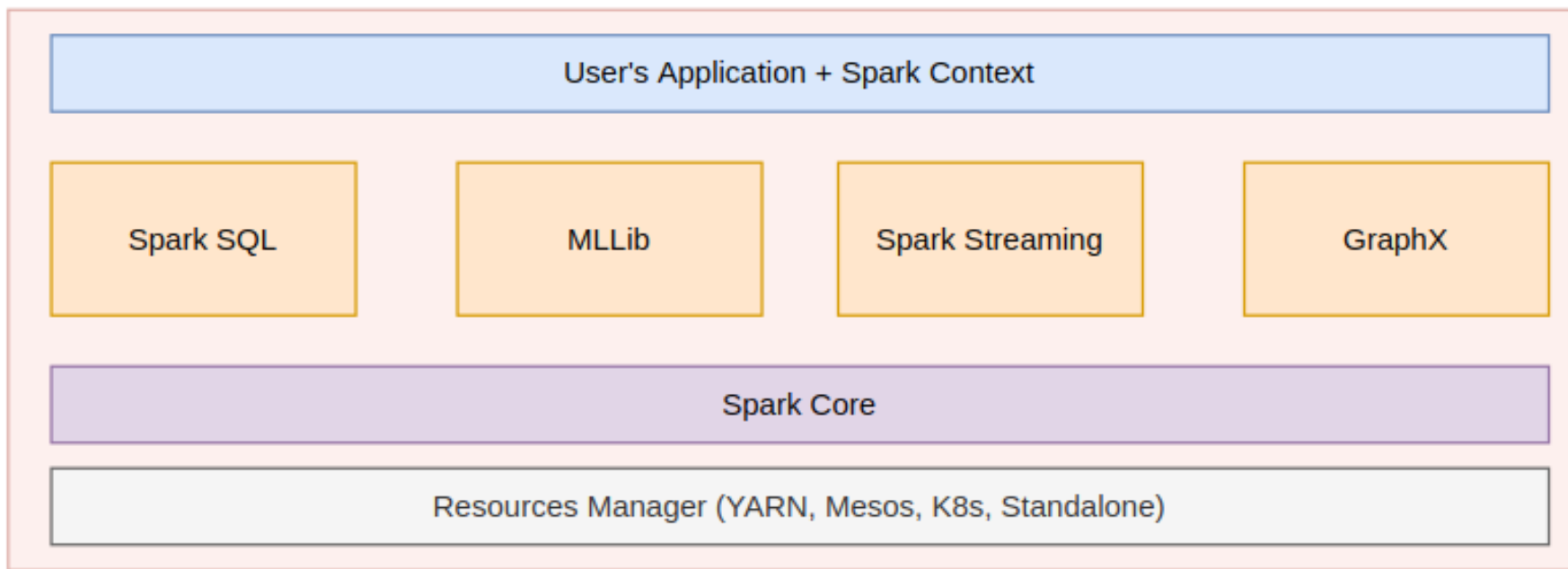
Apache Spark

- Big Data processing Engine (batch and streaming)
- In-memory data pipeline (ETL)
- Parallel execution
- Machine Learning

Spark Architecture



Spark Framework





Scala

- Object-oriented programming language
- Coupled with Java (Java bytecode/Gateway)
- Statically typed / Multithreading
- Spark is built in Scala



Hands-On

- <https://github.com/osekoo/hands-on-spark-scala>
- Preparing dev environment
- Writing code
- Building/testing
- Packaging and submitting tasks





IntelliJ

- Integrated Development Environment
- Multiple modules available for Scala, SBT, Big Data, etc.
- Easy to debug
- Profiler available



Docker

- Platform as a service (PaaS)
- OS abstraction
- Self-contained applications running in containers
- Simplify applications delivery/distribution
- Enable (auto)scaling



Lab Session

- Sync Spark, Hadoop, Scala, Java and SBT versions
- We use here Spark 3.0.2, Hadoop 2.7, Scala 2.12.x, Java 8 and SBT 1.x
- Other details on the Github page



Troubleshooting

- Raise any question during the session or via email kodjo.klouvi@gmail.com